YAKEEN

NEET 2.0 2027

Physical Chemistry | Lecture - 05

REDOX REACTIONS

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Topics to be covered

1. Applications of Redox Reactions
2.
3.
4.

Recap

Ion-electron method
↓
Half rx^n method

Oxidation

Reduction

→ All other atom
→ Balance 'O' by adding H_2O
→ Balance H by adding $H^{\oplus}$

NOTE → In Basic medium, $OH^{\ominus}$ add to both sides

→ Balance charge by adding $e^{\ominus}$

→ Make $e^{\ominus}$ equal and add both half rx^n

Oxidation No. method

Step-I → $\left\{ \begin{array}{l} \text{Inc. in O.N} = \\ \text{dec in O.N} = \end{array} \right\}$ equal

Step-II → Balance charge → $H^{\oplus}$ (Acidic/Neutral)
→ $OH^{\ominus}$ (Basic)

Step-III → Balance 'H & O' by H_2O

Question

NCERT

Permanganate(VII) ion, MnO_4^- in basic solution oxidises iodide ion, I^- to produce molecular iodine (I_2) and manganese (IV) oxide (MnO_2). Write a balanced ionic equation to represent this redox reaction.

Sol:

$$4H_2O + \boxed{2MnO_4^{\ominus} + 6I^{\ominus} \xrightarrow[\text{medium}]{\text{Basic}} 3I_2 + 2MnO_2} + 8OH^{\ominus}$$

$+7$ $\quad$ -1 $\quad$ $\boxed{0}$ $\quad$ $+4$

inc $= 2 \times 3$

dec $= 3 \times 2$

$\boxed{\begin{array}{c} -2-6 \\ \Rightarrow -8 \end{array}}$

$\boxed{\begin{array}{c} 0 \\ -8 \\ \hline -8 \end{array}}$

4 'O' atom + 8 'O' atom

8 'H' atom

12 'O' atom

8 'H' atom

Q How many moles of $I^{\ominus}$ is oxidised to I_2 by using 3 moles of $MnO_4^{\ominus}$ in basic medium?

Sol:

$\because$ 2 mol $MnO_4^{\ominus}$ oxidises 6 moles of $I^{\ominus}$

$\therefore$ 1 ——— $\frac{6}{2}$

$\therefore$ 3 ——— $\frac{6}{2} \times 3 = 9$ mol of $I^{\ominus}$

Question

Balance the following redox reactions:

$MnO_4^-(aq) + SO_2(g) \rightarrow Mn^{2+}(aq) + HSO_4^-(aq)$

(in acidic solution)

Sol:-

$$\boxed{2H_2O + H^{\oplus} + 2MnO_4^{\ominus} + 5SO_2 \longrightarrow 2Mn^{+2} + 5HSO_4^{\ominus}}$$

$+7 \quad +4 \quad +2 \quad +6$

dec = 5 × 2

inc = 2 × 5

$\boxed{-2}$

$+1$

-1

$\boxed{+4-5=-1}$

Question ***

Balance the following redox reactions:

$H_2O_2(aq) + Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + H_2O(l)$ (in acidic solution)

Sol:-

$$2H^{\oplus} + \boxed{H_2O_2 + 2Fe^{+2} \longrightarrow 2Fe^{+3} + 2H_2O}$$

$-1 \qquad +2 \qquad +3 \qquad -2$

inc $= 1 \times 2$

dec $= 2$

$+4$

$+6$

Question

Balance the following redox reactions:

$Cr_2O_7^{2-} + SO_2(g) \rightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$ (in acidic solution)

Sol:-

$$2H^{+} + \boxed{Cr_2O_7^{2-} + 3SO_2 \longrightarrow 2Cr^{+3} + 3SO_4^{-2}} + H_2O$$

+6 +4 +3 +6

dec = 6

inc = 2 $\times$ 3

$$\begin{array}{r} -2 \\ +2 \\ \hline 0 \end{array}$$

+6 − 6 = 0

Question

For the redox reaction,

$$MnO_4^{-} + C_2O_4^{2-} + \boxed{H^{+}} \rightarrow Mn^{2+} + CO_2 + H_2O$$

The correct coefficients of the reactants for the balanced equation are:

	MnO_4^{-}	$C_2O_4^{2-}$	H^{+}
(1)	16	5	2
(2)	2	5	16
(3)	2	16	5
(4)	5	16	2

$$16H^{+} + 2MnO_4^{-} + 5C_2O_4^{2-} \xrightarrow[\text{medium}]{\text{acidic}} 2Mn^{+2} + \overset{10}{\cancel{2}}CO_2 + 8H_2O$$

$+7$ $\quad$ $+3$ $\quad$ $+2$ $\quad$ $+4$

dec $= 5 \times 2$

inc $= 2 \times 5$

-2
-10
-12
$+16$
$+4$

$+4$

Question

In the half reaction: $2ClO_3^- \rightarrow Cl_2$

1. ~~5 electrons are gained~~
2. ~~5 electrons are liberated~~
3. 10 electrons are gained
4. ~~10 electrons are liberated~~

$\overset{+5}{2ClO_3^-} + 10e^- \longrightarrow \overset{0}{Cl_2}$

Reduction

Question

The number of electrons required to balance the following equation–
$NO_3^- + 4H^+ + \boxed{e^-} \rightarrow 2H_2O + NO$ are–

(1) 5

(2) 4

(3) 3

(4) 2

$$\boxed{3e^{\ominus} + \overset{+5}{NO_3^{\ominus}} \longrightarrow \overset{+2}{NO}}$$

Reduction half rxn

$$\boxed{3e^{\ominus} + 4H^{\oplus} + NO_3^{\ominus} \longrightarrow NO + 2H_2O}$$

$+4$
-1
$+3$
-3
0

0

Question

$C_2H_6(g) + \frac{7}{2}\cancel{n}O_2 \rightarrow 2CO_2(g) + 3H_2O(l)$

In this equation, the ratio of the coefficients of CO_2 and H_2O are

1. $1:1$
2. $2:3$
3. $3:2$
4. $1:3$

Question

dec in O.N = 10

$+7 \quad -1 \quad Mn^{+x} \quad 0$

$2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Z + 5O_2 + 8H_2O$. In this reaction Z is:

inc. in O.N = 10

1. Mn^{+2}
2. Mn^{+4}
3. MnO_2
4. Mn

$\cancel{2}(7-x) = \cancel{10}^{5} = \text{dec. in O.N}$

$7 - x = 5$

$x = +2$

Question

inc in O.N = 2

What is 'A' in the following reaction $2Fe^{3+}_{(aq)} + Sn^{2+}_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + A$

Sn^{+4}

dec. in O.N = 2

1. $Sn^{3+}_{(aq)}$
2. $Sn^{4+}_{(aq)}$ ✓
3. $Sn^{2+}_{(aq)}$
4. Sn

Balancing of disproportionation rxn / Comproportionation rxn

→ follow the same rules but use common sense while balancing single reactant or product undergoing disproportionation or Comproportionation

Disproportionation

$$2OH^{\ominus} + Cl_2 \xrightarrow[\text{medium}]{\text{Basic}} Cl^{\ominus} + ClO^{\ominus} + H_2O$$

$0 \qquad -1 \qquad +1$

Oxidation no.

dec = 1

inc = 1

$+3H_2O$

$$3OH^{\ominus} + P_4 \xrightarrow[\text{medium}]{\text{Basic}} PH_3 + 3H_2PO_2^{\ominus}$$

$0 \qquad -3 \qquad +1$

Oxidation no.

dec = 3

inc = 1 × 3

Comproportionation

$$5Br^{\ominus} + BrO_3^{\ominus} \longrightarrow 3Br_2$$

$-1 \qquad +5 \qquad 0$

dec = 5

inc = 1 × 5

$$6OH^{\ominus} + 3Cl_2 \xrightarrow[\text{medium}]{\text{Basic}} 5Cl^{\ominus} + ClO_3^{\ominus} + 3H_2O$$

$0 \qquad -1 \qquad +5$

dec = 1 × 5

inc = 5

Applications of redox reactions

(1) n-factor calculation

(2) Equivalent mass = $\frac{\text{Molar mass}}{\text{n-factor}}$

(3) Law of equivalence

(4) Titration

acid-Base titration

Redox titration

(1) N-factor calculation & Equivalent Mass

Oxidation half rxn $\Rightarrow$ $\boxed{\text{n-factor = inc. in O.N or no. of } e^{\ominus} \text{ lost}}$

$Fe^{+2} \longrightarrow Fe^{+3} + e^{\ominus}$

n-factor = 1

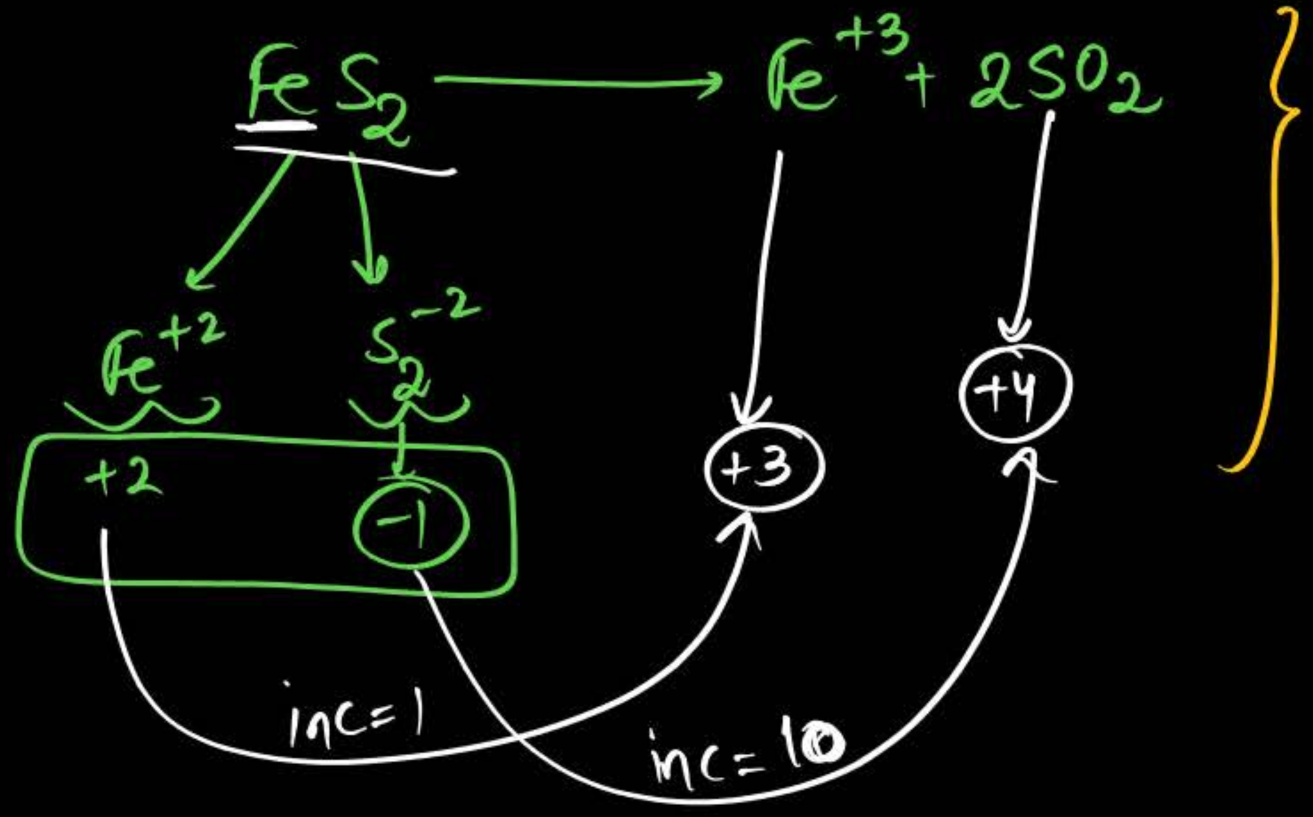
$Cu \longrightarrow Cu^{+2} + 2e^{\ominus}$

n-factor = 2

inc. in O.N = 2

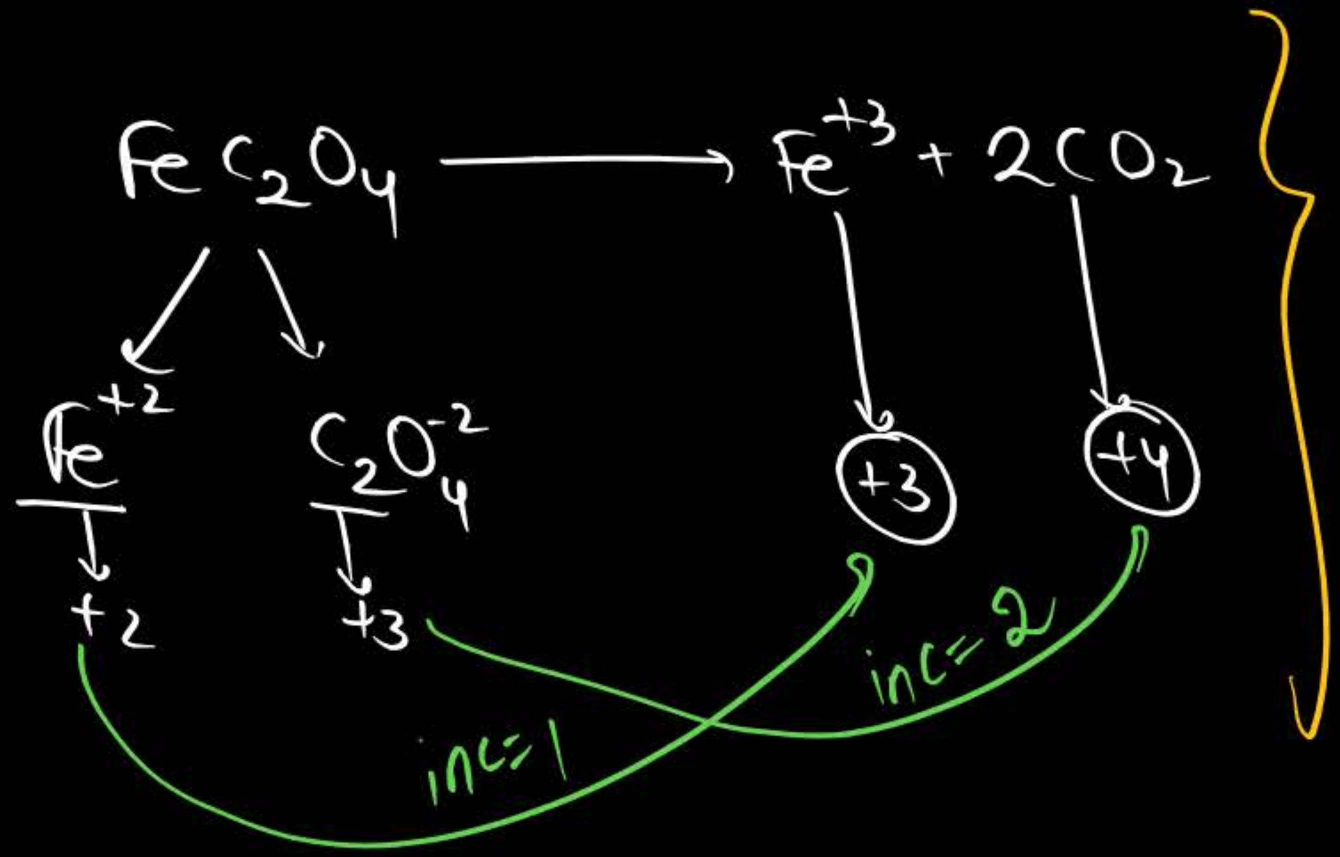
$\overset{+3}{C_2}O_4^{-2} \longrightarrow 2\overset{+4}{C}O_2$

$E_{C_2O_4^{-2}} = \frac{M_{C_2O_4^{-2}}}{2}$



n-factor for $FeS_2 = 1 + 10 = 11$

$$E_{FeS_2} = \frac{M_{FeS_2}}{11}$$



n-factor for $FeC_2O_4 = 1 + 2 = 3$

$$E_{FeC_2O_4} = \frac{M_{FeC_2O_4}}{3}$$

$\Rightarrow$ Reduction half rxn $\rightarrow$ $\boxed{\text{n-factor = dec. in O.N or no. of } e^{\ominus} \text{ gained}}$

$$Cl_2 + 2e^{\ominus} \longrightarrow 2Cl^{\ominus}$$

$\hookrightarrow$ n-factor = 2

$\hookrightarrow E_{Cl_2} = \dfrac{M_{Cl_2}}{2}$

$$6e^{\ominus} + \overset{+6}{Cr_2O_7^{2-}} \longrightarrow 2Cr^{+3}$$

$\hookrightarrow$ n-factor for $Cr_2O_7^{2-} = 6$

$\hookrightarrow E_{Cr_2O_7^{2-}} \Longrightarrow \dfrac{M_{Cr_2O_7^{2-}}}{6}$

$$KMnO_4^{\ominus} \text{ or } \overset{+7}{MnO_4^{\ominus}} + 5e^{\ominus} \xrightarrow{\text{acidic medium}} Mn^{+2}$$

n-factor for $KMnO_4 = 5$

$E_{KMnO_4} = \dfrac{M_{KMnO_4}}{5}$

$$KMnO_4 \text{ or } \overset{+7}{MnO_4^{\ominus}} + 3e^{\ominus} \xrightarrow[\text{Neutral medium}]{\text{slightly alkaline or}} \overset{+4}{MnO_2}$$

$\rightarrow$ n factor for $KMnO_4 = 3$

$E_{KMnO_4} = \dfrac{M_{KMnO_4}}{3}$

$$KMnO_4 \text{ or } \overset{+7}{MnO_4^{\ominus}} + 1e^{\ominus} \xrightarrow[\text{medium}]{\text{Basic/alkaline}} \overset{+6}{MnO_4^{-2}}$$

$\rightarrow$ n-factor for $KMnO_4 = 1$

$E_{KMnO_4} = \dfrac{M_{KMnO_4}}{1}$

Q Calculate Eq. mass of H_2O_2 as it undergoes $\boxed{H_2O_2 \xrightarrow[\text{medium}]{\text{acidic}} 2H^{\oplus} + O_{2(g)}\uparrow + 2e^{\ominus}}$.

Sol:-

Molar mass of $H_2O_2 = 1\times2+16\times2 = 34\,g$

$$E_{H_2O_2} = \frac{M_{H_2O_2}}{2} = \frac{34}{2} = 17\,g$$

$\Rightarrow$ $\boxed{\text{Redox Rx}^n \text{ (Balanced)}}$ $\rightarrow$ $\boxed{\text{n-factor} = (\text{Inc. in O.N} = \text{dec. in O.N.})}$

Reduction half rxn: $\overset{+7}{MnO_4^{\ominus}} \longrightarrow \overset{+4}{MnO_2}$ (n-factor = 3)

Oxidation half rxn: $2I^{\ominus} \longrightarrow I_2$ (n-factor = 2)

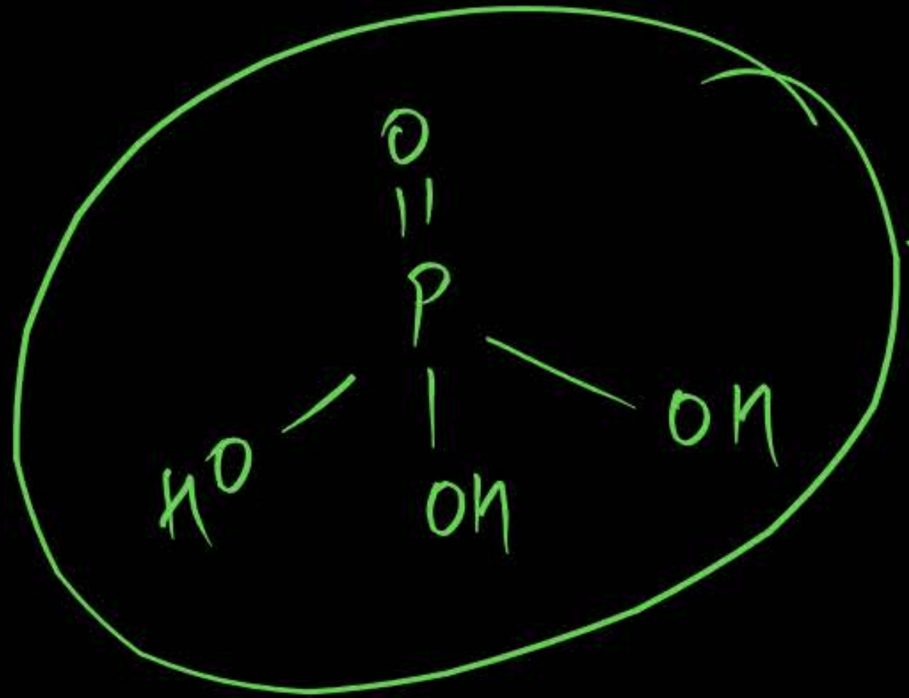
Balanced redox rxn $\rightarrow$ $\boxed{4H_2O + 2MnO_4^{\ominus} + 6I^{\ominus} \longrightarrow 3I_2 + 2MnO_2 + 8OH^{\ominus}}$ $\rightarrow$ $\boxed{\text{n-factor} = 6}$

Question

What is the n-factor for H_3PO_4 in the reaction $2NaOH + \underline{H_3PO_4} \rightarrow Na_2HPO_4 + 2H_2O$?

n-factor = 2

$$E_{H_3PO_4} = \frac{M_{H_3PO_4}}{2}$$



n factor = 3

Question

Calculate the n-factor of reactants in the given chemical changes?

1. $\overset{+6}{K_2Cr_2O_7} \xrightarrow{H^+} 2Cr^{+3}$ → n-factor = 6

2. $\overset{+3}{C_2O_4^{2-}} \rightarrow 2\overset{+4}{CO_2}$ → n-factor = 2

3. $\overset{+2}{S_2O_3^{2-}} \xrightarrow{\text{alkaline}} 2\overset{+6}{SO_4^{2-}}$ → n-factor = 8

4. $\overset{-1}{I^-} \longrightarrow \overset{+1}{ICl}$ → n-factor = 2

$$\overset{0}{N_2} \longrightarrow \overset{-3}{NH_3}$$

n-factor for $N_2 = 6$

$$E_{N_2} = \frac{M_{N_2}}{6} = \frac{28}{6}$$

$$\overset{-3}{NH_3} \longrightarrow \overset{0}{N_2}$$

n-factor for $NH_3 = 3$

$$E_{NH_3} = \frac{M_{NH_3}}{3} = \frac{17}{3}$$

$$\overset{-3}{NH_3} \longrightarrow \overset{+5}{HNO_3}$$

n-factor for $NH_3 = 8$

$$E_{NH_3} = \frac{17}{8}$$

$$\overset{+5}{HNO_3} \longrightarrow \overset{-3}{NH_3}$$

n-factor for $HNO_3 = 8$

$$E_{HNO_3} = \frac{M}{8}$$

Homework

Attempt DPP-04

Thank You